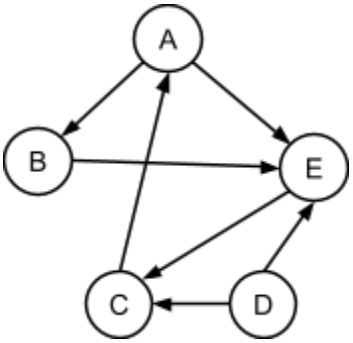
American Computer Science League

2024-2025 ● Contest 4: Short Problems ● Intermediate Division

1. Graph Theory

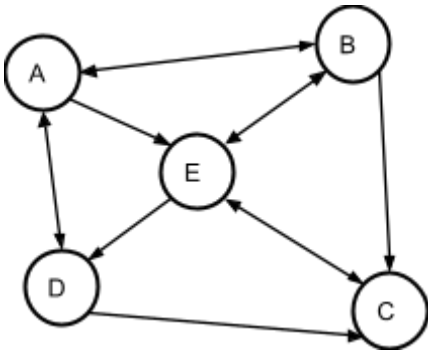
How many paths of length 2 are there in the following directed graph?



A. 6
B. 7
C. 8
D. 9
E. 10

2. Graph Theory

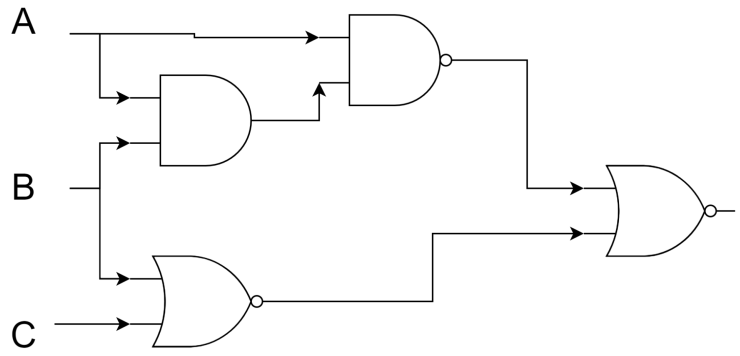
How many cycles are there in this directed graph?



A. 8
B. 9
C. 10
D. 11
E. 12

3. Digital Electronics

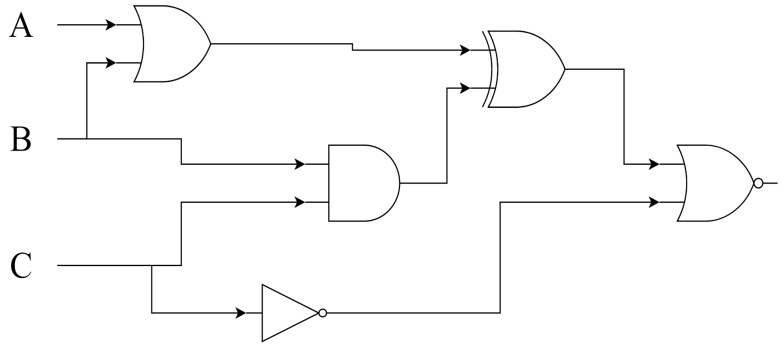
How many ordered triples make the following digital circuit TRUE?



A. 0
B. 2
C. 4
D. 6
E. 8

4. Digital Electronics

Translate the following digital circuit to a Boolean algebra expression.



A. $\overline{AB \oplus (B + C) + \overline{C}}$
B. $\overline{(A + B) \oplus BC} + \overline{C}$
C. $\overline{A + B \oplus BC + \overline{C}}$
D. $\overline{(A + B) \oplus BC + C}$
E. $(A + B) \oplus BC + \overline{\overline{C}}$

5. Assembly Language

What is the output when the following program is executed?

```
NUM     DC      2025
        LOAD    NUM
        DIV     =100
        STORE   A
        MULT    =100
        SUB     NUM
        MULT    =-1
        STORE   B
        DIV     =5
        MULT    =10
        STORE   ANS
        LOAD    A
        DIV     =5
        ADD     ANS
        STORE   ANS
        PRINT   ANS
        END
```

A. 45
B. 54
C. 405
D. 504
E. 99

6. Assembly Language

What is the output when the following program is executed?

```
T       DC      0
N       DC      19
NUM     DC      135
TOP     LOAD    NUM
        SUB     T
        BE      STOP
        LOAD    N
        ADD     =1
        STORE   N
        LOAD    T
        ADD     N
        STORE   T
        BU      TOP
STOP    PRINT   N
        END
```

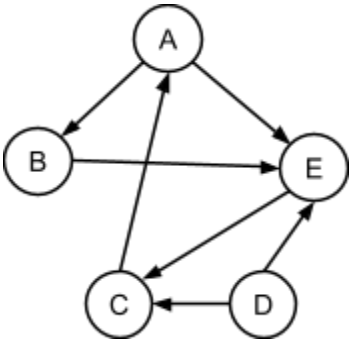
A. 21
B. 22
C. 23
D. 24
E. 25

American Computer Science League

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1. 图论

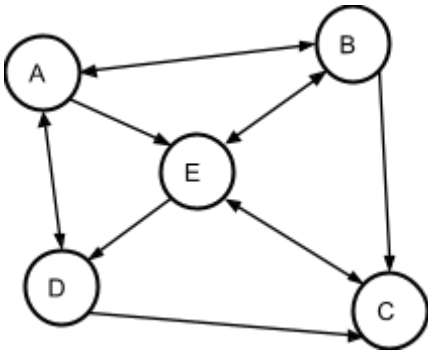
请问在以下有向图中，长度为 2 的路径有多少条?



A. 6
B. 7
C. 8
D. 9
E. 10

2. 图论

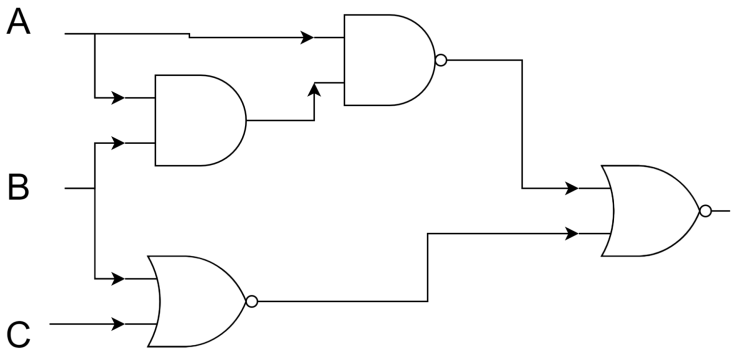
请问在以下有向图中存在多少个环？



A. 8
B. 9
C. 10
D. 11
E. 12

3. 数字电路

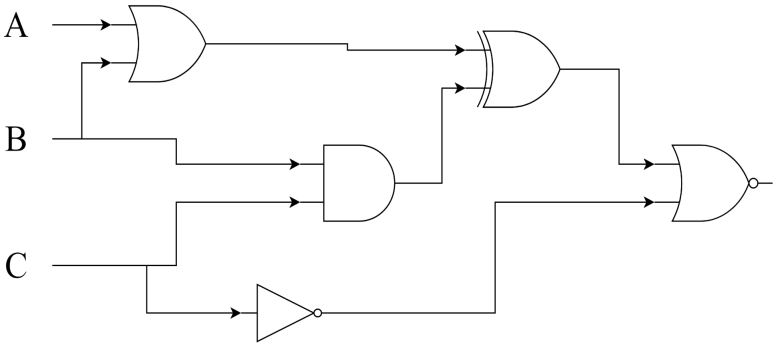
请问有多少个有序三元组可以使得下述数字电路为**真**？



A. 0
B. 2
C. 4
D. 6
E. 8

4. 数字电路

请将以下数字电路转换为布尔代数表达式。



A. $\overline{AB \oplus (B + C) + \overline{\overline{C}}}$
B. $\overline{(A + B) \oplus BC} + \overline{\overline{C}}$
C. $\overline{A + B \oplus BC + \overline{\overline{C}}}$
D. $\overline{(A + B) \oplus BC + C}$
E. $(A + B) \oplus BC + \overline{\overline{C}}$

5. 汇编语言

请问执行下列程序时会输出什么？

```
NUM     DC      2025
        LOAD    NUM
        DIV     =100
        STORE   A
        MULT    =100
        SUB     NUM
        MULT    =-1
        STORE   B
        DIV     =5
        MULT    =10
        STORE   ANS
        LOAD    A
        DIV     =5
        ADD     ANS
        STORE   ANS
        PRINT   ANS
        END
```

A. 45
B. 54
C. 405
D. 504
E. 99

6. 汇编语言

请问执行下列程序时会输出什么？

```
T       DC      0
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NUM     DC      135
TOP     LOAD    NUM
        SUB     T
        BE      STOP
        LOAD    N
        ADD     =1
        STORE   N
        LOAD    T
        ADD     N
        STORE   T
        BU      TOP
STOP    PRINT   N
        END
```

A. 21
B. 22
C. 23
D. 24
E. 25